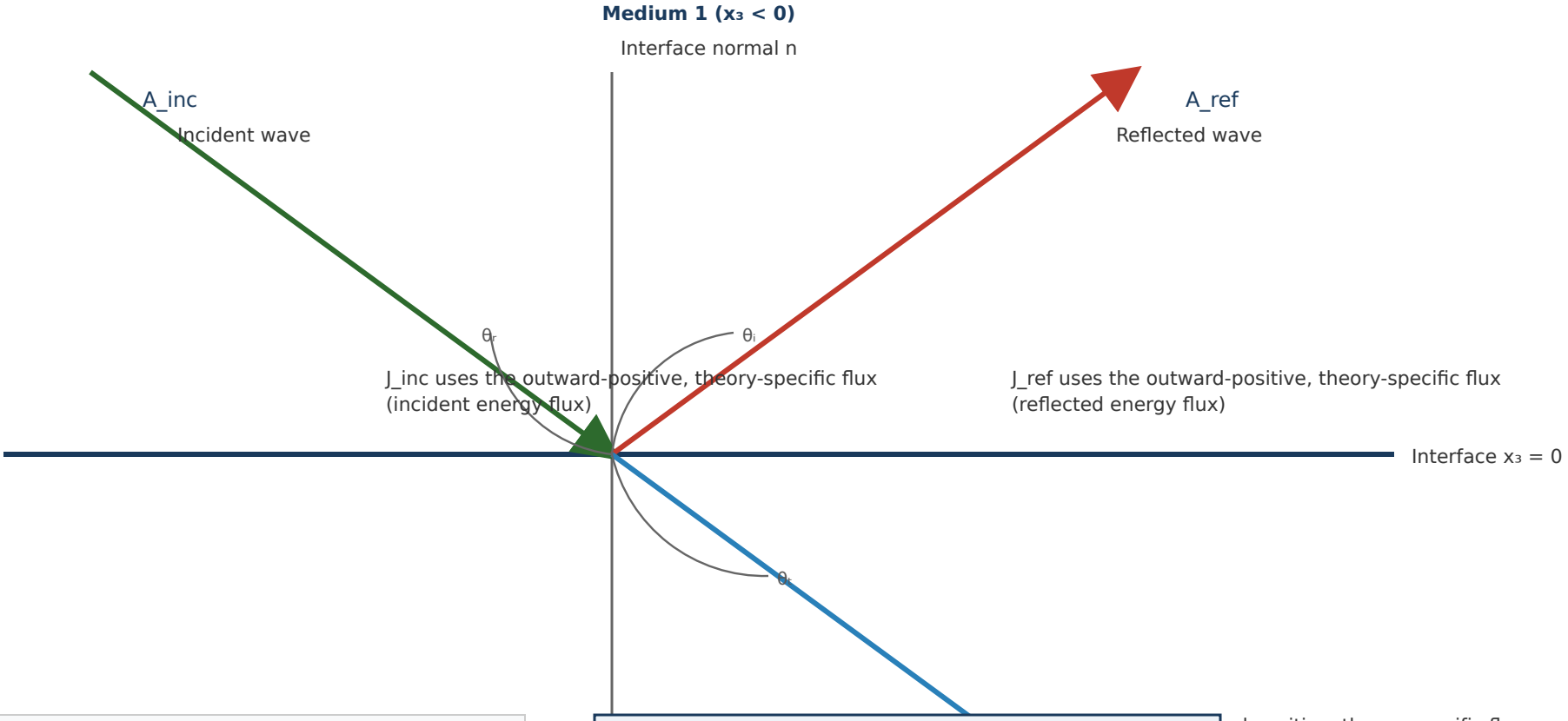
Interface reflection and transmission

Incident, reflected, and transmitted waves at a bonded interface between two media



Interface conditions (perfect bonding):

$u^{(1)}_i = u^{(2)}_i$ (displacement continuity)

$\sigma^{(1)}_{ji} n_j = \sigma^{(2)}_{ji} n_j$ (traction equilibrium)

$T^{(1)} = T^{(2)}$ (temperature continuity)

$q^{(1)}_i n_i = q^{(2)}_i n_i$ (heat flux continuity)

Energy conservation:

$R + T + A = 1$

R: reflected energy fraction

T: transmitted energy fraction

A: absorbed (dissipated) fraction

conditional on complete flux, passivity, and stated interface storage

ard-positive, theory-specific flux
ux)

For generalized theories, en

Benchmark status is not inde

rans

d wave